ERYTHROMYCIN

QT prolongation
CYP3A4 inhibitor

CHEMICAL NAME (3R,4S,5S,6R,7R,9R,11R,12R,13S,14R)-6-[(2S,3R,4S,6R)-4-(dimethylamino)-3-hydroxy-6-methyloxan-2-yl]oxy-14-ethyl-7,12,13-trihydroxy-4-[(2R,4R,5S,6S)-5-hydroxy-4-methoxy-4,6-dimethyloxan-2-yl]oxy-3,5,7,9,11,13-hexamethyl-oxacyclotetradecane-2,10-dione

CLASS Macrolide antibiotic; motilin agonist

TRADE NAMES Erythrocin · Erymax · E-Mycin · Ery-Ped

THERAPEUTIC USES / INDICATIONS

- Respiratory and skin infections in penicillin-allergic patients.
- Pertussis; diphtheria (carriers); *Chlamydia* in pregnancy; acne.
- Gastroparesis and to empty stomach before endoscopy (prokinetic, low dose).
- Neonatal conjunctivitis prophylaxis (eye ointment).

ADMINISTRATION

- **PO:** base tablets 1 h before or 2 h after food (stearate); ethylsuccinate with or without food.
- **IV:** infusion over 60 min (diluted) — painful, use large vein; never IM.
- Eye ointment for neonates.

DOSAGES

- **Adult:** 250–500 mg every 6 h, or 500 mg–1 g every 12 h; max 4 g/day.
- **Children:** 30–50 mg/kg/day divided every 6–8 h.
- **Prokinetic (off-label):** single IV dose before endoscopy in severe upper GI bleeding (ESGE).
- **Acne:** 250–500 mg twice daily.

PHARMACOKINETICS

- **Peak:** 1–4 h. Destroyed by gastric acid (enteric-coated forms used).
- Metabolised in liver (CYP3A4); concentrated in bile.
- Excreted in bile and faeces. **t½:** 1.5–2 h.

PHARMACODYNAMICS

- Binds **50S ribosome** → bacteriostatic protein-synthesis inhibition.
- Stimulates **motilin receptors** → ↑ gastric emptying (cause of GI side effects).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting, abdominal cramps, diarrhoea (very common).
- Rash.
- Thrombophlebitis with IV.

ADVERSE EFFECTS (SERIOUS)

- QT prolongation, torsades de pointes.
- Cholestatic hepatitis (estolate salt).
- Ototoxicity (reversible, high IV doses).
- Infantile hypertrophic pyloric stenosis (neonates <2 weeks).

CONTRAINDICATIONS & CAUTIONS

- Macrolide hypersensitivity.
- Use with astemizole, terfenadine, pimozide, ergotamine, simvastatin, colchicine.
- Long QT; hypokalaemia.
- **Caution:** liver disease, myasthenia gravis.

KEY INTERACTIONS

- **CYP3A4 substrates:** ↑ levels (statins, carbamazepine, theophylline, ciclosporin, warfarin, midazolam).
- **QT-prolonging drugs:** additive.

NURSING MANAGEMENT

- Give with plenty of water at correct time relative to meals.
- IV: dilute adequately; slow infusion; check site.
- Monitor ECG in high-risk patients.
- Monitor GI tolerance, LFTs.
- Check interacting medications.

PATIENT EDUCATION

- Stomach upset is common; take as directed.
- Report palpitations, yellow skin, hearing problems.
- Complete the course.

KEY POINTS

- Oldest macrolide; many GI side effects (motilin agonist).
- QT prolongation; CYP3A4 inhibitor.
- Prokinetic use for gastroparesis.
- Pyloric stenosis risk in neonates.

SOURCES CHECKED **[159]** Erythromycin 125 mg/5 ml Sugar Free Granules for Oral Suspension (UK SmPC) **[160]** Erythromycin 1g Powder for Solution for Infusion (UK SmPC) **[161]** Erythromycin, oral (US label) **[162]** Erythromycin, oral (US label, Torrent Pharmaceuticals Limited)
Full references p. 432–457